What Is the Alpha on Polymarket? A Methodology of Probability Mispricing and a Taxonomy of Eleven Profitable Regimes

Empirical Companion Paper

Wallet-level data: `polydata.pro`, snapshot of 21 April 2026

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Abstract

A companion to *"Smart Money on Polymarket"* (Paper I), this paper asks a sharper, methodological question: *what is the alpha?* We argue and document that every profitable wallet in our $N = 273$ panel is running the **same** underlying strategy—*probability mispricing arbitrage*—differing only in (i) which part of the price spectrum the mispricing lives, and (ii) what generates it. We formalise the unifying framework, decompose the eleven empirical regimes documented in Paper I onto four **alpha sources**—slow information assimilation, behavioural biases, microstructure inefficiency, and cross-market inconsistency—and show that aggregate profits map cleanly onto these sources. The *slow-information* bucket alone accounts for ∼\$140M of the \$177M total realised PnL in our panel, dominated by Politics specialists holding under-priced high-probability YES shares to a \$1 resolution payoff. Behavioural-bias arbitrage (long-shot, contrarian) is small in absolute PnL but the *highest-Sharpe* bucket. Microstructure (HFT/MM) and cross-market (sports model, weather model) buckets contribute moderately. The single useful insight for a practitioner is unambiguous: **Polymarket is not a casino and not a high-frequency arbitrage venue; it is a venue where positive expected value comes exclusively from being correctly contrarian to a measurable mispricing in the implied-probability surface.**

1 Introduction: One Question

Paper I documented that 273 wallets analysed by polydata.pro generated \$177 M of realised PnL on \$9.5 B of cumulative volume; that profits are extremely concentrated (top decile = 52%); that 67% of those wallets are at least partly automated; and that the dominant signature is a 93% event win-rate paired with a 58% trade win-rate—a classic "DCA into a binary view, hold to resolution" fingerprint.

The question this paper answers is sharper: *What, exactly, is the edge?* If 11 distinguishable strategies all make money, are they all exploiting different things, or are they manifestations of a common phenomenon? And whatever it is—can we name it precisely enough to write down a P&L formula?

The answer we develop is:

> *Every profitable Polymarket wallet is, mechanically, executing probability mispricing arbitrage. The expected value of any trade is* $(p_{true} - p_{market}) \times \$1 - c$*, where* c *collects fees, slippage and adverse-selection.* All 11 regimes exist because the spread $p_{\text{true}} - p_{\text{market}}$ has exactly four economically-meaningful sources, and each regime is a specific operational way of locating one of them.

The paper's structure mirrors that claim. Section 2 writes the trade-level expectation. Section 3 names the four alpha sources. Section 4 maps each of the 11 empirical regimes onto a source and presents the per-regime numbers that operationalise the mapping. Section 5 aggregates PnL by source. Section 6 draws the implications, and Section 7 closes.

2 The Trade-Level Identity

2.1 Notation

A Polymarket binary contract pays \$1 on resolution if the event occurs and \$0 if it does not. Let $p^* \in [0,1]$ be the unknown true probability of the event and $p_m \in (0,1)$ the current market price for one YES share (so equivalently $1 - p_m$ is the NO share price; YES + NO must sum to \$1 by construction). Let q be the quantity (shares) of YES bought. The expected one-shot gross return is

$$\mathbb{E}[\Pi_{\text{gross}}] \;=\; q \cdot \big(p^* \cdot 1 + (1-p^*) \cdot 0 - p_m\big) \;=\; q \cdot (p^* - p_m). \tag{1}$$

Subtracting microstructure costs (gas, exchange fee, half-spread, slippage) collected as $c \cdot q$,

$$\mathbb{E}[\Pi] \;=\; q \cdot \big(p^* - p_m - c\big). \tag{2}$$

The trade has positive expected value iff $p^* > p_m + c$. Symmetrically, buying NO has positive EV iff $p^* < (1 - p_m) - c$, i.e. $p_m > p^* + c$. *Both directions of the same inequality are mispricing arbitrage; only the sign differs.*

2.2 Implications of (2)

Three properties of (2) drive the entire empirical landscape of Paper I.

(i) The edge is multiplicative in q. Position size q amplifies any mispricing $p^* - p_m$. This is why *capital matters*: a 5 ¢mispricing on \$10 k of stake is \$500; on \$1 M of stake it is \$50 k.

(ii) The microstructure cost c has a floor and a slope. A wallet that turns over rapidly accumulates many copies of c; a wallet that buys once and waits for resolution pays c once. Hence DCA-into-resolution dominates HFT for any given $|p^* - p_m|$, all else equal.

(iii) Mispricing must come from *somewhere*. The market price p_m is set by the marginal counter-party. A persistent $|p^* - p_m| > c$ requires either a slow updater (information mispricing), a biased updater (behavioural mispricing), a microstructure inefficiency (bid–ask wider than fair-value uncertainty), or a cross-market inconsistency (the same outcome priced differently in two venues or two contracts). These four are the alpha sources.

2.3 From single-trade EV to portfolio outcome

Most regimes execute many trades over time on the same underlying. Let the operator place T trades indexed $t = 1, \ldots, T$ on a single event, each at price $p_{m,t}$ and quantity q_t. The aggregate event-level PnL is

$$\Pi_{\text{event}} \;=\; \mathbf{1}\{\text{event occurs}\} \cdot Q - \sum_{t=1}^{T} q_t p_{m,t} - cQ, \quad \text{where } Q = \sum_t q_t. \tag{3}$$

$\mathbb{E}[\Pi_{\text{event}}] = (p^* - \bar{p}_m - c)Q$ with $\bar{p}_m = \sum_t q_t p_{m,t}/Q$ the volume-weighted average entry price. *DCA wins when* $\bar{p}_m < p^*$; this is mathematically identical to the trade-level rule but applied to volume-weighted entries. It explains why 93% *event* win-rates can coexist with 58% *trade* win-rates: each individual $p_{m,t}$ might exceed p^* momentarily (so that selling immediately after would lose money), but the average $\bar{p}_m$ is well below p^*.

3 The Four Alpha Sources

3.1 S1. Slow information assimilation (informational alpha)

Public information that ought to be in p_m is not yet, because the marginal counter-party has not seen, processed, or believed it. The classic example: a poll dropping at 8:00 AM that moves the true probability of a senate seat from 55% to 70% is reflected in p_m over hours, not seconds. Operators reading the poll first ("information sniping") and operators with a calibrated long-run prior ("DCA holders") both arbitrage S1, just on different time horizons.

3.2 S2. Behavioural biases

Counter-parties price systematically away from p^* for psychological reasons that survive learning. Two are diagnostic on Polymarket:

- **Long-shot bias**: low-probability events are *under-priced* (the opposite of the racetrack favourite-long-shot effect of Griffith, 1949, Thaler and Ziemba, 1988, Snowberg and Wolfers, 2010, because retail prediction-market participants overestimate "boring" favourites and under-stake long-shots). The lottery regime arbitrages this by buying many sub-20 ¢YES shares.
- **Recency / partisan momentum**: prices over-react to the most recent news or social-media signal. The contrarian regime arbitrages this by entering the opposite side of any extreme price.

3.3 S3. Microstructure inefficiency

The bid-ask spread on Polymarket is wider than the residual fair-value uncertainty, especially in low-volume markets. A passive liquidity provider who simultaneously quotes both sides at $p \pm \epsilon$ collects 2ϵ per round-trip with positive expected return so long as adverse selection is bounded. This is operationally equivalent to traditional market-making and is the domain of the strict HFT regime.

3.4 S4. Cross-market / cross-venue inconsistency

Two facts about the same world should imply equal prices. When they do not, you can lock in a positive expected return by trading the mispriced leg. Examples:

- Polymarket's NBA spread on Lakers -7.5 implies an underlying probability that may differ from Vegas's number on the same line.
- Polymarket's binary "BTC $>$ \$70 k by Friday" implies a log-normal volatility that may differ from Deribit's option chain.
- Polymarket's binary "Miami high $>$ 88°F tomorrow" should match the ECMWF ensemble forecast.

4 Mapping Regimes to Sources

Table 1 consolidates every regime catalogued in Paper I against the four alpha sources, with the per-regime statistical fingerprint that makes the mapping operational rather than rhetorical. We discuss each row in turn; the underlying numbers are computed from our $N = 273$ panel.

Table 1: The eleven empirical regimes mapped to the four alpha sources. Per-regime statistics are medians across the wallets in each bucket. "<20" / "mid" / ">55" = % of position size in the <20¢ 20–55¢ >55¢probability bands. "YES" = median YES position share. "Sell" = % of trades that exit by selling rather than redeeming at resolution. "DCA" = average entries per market. "% win" = share of wallets in the bucket with positive PnL.

#	Regime	n	YES	<20	mid	>55	Sell	DCA	Med PnL	% win
R1	Politics DCA bot/semi-bot [S1]	33	46	3	14	**68**	28	72	$591k	100
R2	Sports spread model bot [S4]	41	40	6	47	45	0	9	$4k	63
R3	Multi-sport scanner bot [S1+S4]	5	45	12	47	45	0	38	$381k	100
R4	Crypto short-horizon bot [S2]	53	**49**	6	30	59	4	9	$34k	79
R5	Politics info-edge human [S1]	30	45	6	14	**67**	24	13	$118k	97
R6	News-driven sniper [S1 timing]	96	44	8	14	70	31	16	$86k	92
R7	Lottery / long-shot [S2]	36	56	**51**	19	25	23	19	$116k	92
R8	Contrarian / mean-reversion [S2]	89	52	19	47	31	1	21	$62k	75
R9	Weather statistical model [S4]	15	56	21	10	42	5	6	$55k	87
R10	Strict HFT / market maker [S3]	31	**50**	13	47	38	0	49	$112k	77
R11	Diamond Hands hold-to-resolution [S1]	78	46	7	47	39	0	18	$51k	76

4.1 R1. Politics DCA bot/semi-bot [S1]

Bet: >55¢shares (median 68% of position size in the high band) on long-resolving political events. **Why $p^* > p_m$:** the marginal counter-party is partisan-retail who reacts to news rather than to baseline polling. The well-calibrated operator buys at 65 ¢when integrated polling implies 80%. **Operational signature:** median DCA 72 entries/market; 28% sell rate (most positions held to resolution); 95% event win-rate. **Realised:** $70.6 M aggregate PnL across 33 wallets, including the panel's two largest profit producers (Theo4, $22 M; Fredi9999, $16.6 M).

4.2 R2. Sports spread bot [S4]

Bet: mid-band (20–55 ¢47% of position size) point-spread / over-under contracts on individual matches. **Why $p^* > p_m$:** Polymarket spreads update slower than Vegas/Pinnacle; statistical models (Elo, regression) detect when the binary contract diverges from the implied probability of the same line elsewhere. **Operational signature:** median DCA 9; sell rate 0% (held to game end, then redeemed); short trade gaps ($\sim$30 s) during the pre-game window. **Example:** Motherly-Insect's three largest wins were Clippers −9.5 ($204 k profit, 99.9% ROI), Warriors −12.5 (97.5% ROI), Thunder −19.5 (96.1% ROI)—all purchased near 50 ¢and redeemed at $1.

4.3 R3. Multi-sport scanner bot [S1+S4]

Bet: thousands of small positions across global sport calendars. **Why $p^* > p_m$:** a low-edge but very-high-volume strategy that aggregates many small mispricings of type S4 plus news-driven S1 (e.g., line moves from injury news). **Operational signature:** >1000 trades/day, >2700

markets touched, 0% sell rate. The breadth diversifies away outcome-specific risk and realises the law-of-large-numbers benefit of a small expected edge.

4.4 R4. Crypto short-horizon bot [S2]

Bet: essentially neutral YES/NO (49% YES bias), *tilted toward* >55*¢shares* (59% of position size). The dominant market type is short-window "Up or Down" contracts (15 min–4 h, BTC/ETH/XRP/SOL). **Why** $p^* > p_m$**:** *this is the single most surprising regime in the panel.* The bot is *not* directional—it is harvesting a tail-mispricing phenomenon. The top wins are dominated by extreme ROIs ($>1000\%$) on contracts whose entry price was below \$0.05—i.e., events the market priced at $<5\%$ but which the operator's model assigned materially higher probability. The net YES-bias near 50% confirms that the bot harvests this on *both* sides of every binary. **Best example:** f705fa04 paid \$68 to enter "BTC > \$68 k on March 27"—a contract priced near zero—and collected \$53 938 (78 690% ROI). The same wallet records dozens of such "zero-priced YES" wins. This is operationally equivalent to *being long convexity at near-zero premium* when the market under-prices the right-tail of crypto price moves.

4.5 R5. Politics info-edge human [S1]

The same regime as R1 with manual DCA. Operational signature is identical except that average DCA per market collapses from 72 to 13—humans cannot DCA at machine cadence. The information edge is identical. **Top wallet:** GCottrell93 (\$3.4 M), whose largest single win was \$158 k on a \$38 k entry into "US x Iran ceasefire by April 15?"—a contract whose true probability the operator clearly priced higher than the market did at the time of entry.

4.6 R6. News-driven sniper [S1, timing]

Bet: essentially neutral; opportunistic buys on either side within minutes of a news event. **Why** $p^* > p_m$**:** the public update has not yet propagated to p_m. The sniper's 31% sell rate (versus 0–5% for DCA holders) is the diagnostic feature: snipers exit *into* the post-news repricing rather than holding to resolution. **Risk:** adverse selection if the news has been front-run. Operates best on macro releases (CPI, jobs, Fed) with scheduled timing.

4.7 R7. Lottery / long-shot [S2]

Bet: 51% of position size in <20 ¢shares; modest YES bias (56%). **Why** $p^* > p_m$**:** the *reverse* of the racetrack favourite-long-shot effect. On Polymarket the long-shot is *under*-priced because retail participants under-stake events that "feel impossible" (Snowberg and Wolfers, 2010, Thaler and Ziemba, 1988 for the analogous racetrack mechanism). A 5 ¢contract whose true probability is 8% has expected return $(0.08 - 0.05)/0.05 = 60\%$ per trade. **Empirical:** 92% profitable, median PnL \$116 k—higher than several "smarter"-looking regimes.

4.8 R8. Contrarian / mean-reversion [S2]

Bet: mid-band concentrated (47% of position size in 20–55 ¢). **Why** $p^* > p_m$**:** when the price is pushed to an extreme by a momentum cascade, p_m over-shoots p^*. The contrarian buys the over-shot side (sells the overpriced YES, buys the underpriced NO, or vice versa). **Empirical:** 75% profitable; lower hit-rate than the average regime, consistent with the operator betting against momentum that sometimes continues.

4.9 R9. Weather statistical model [S4]

Bet: both tails (21% in <20¢ 42% in >55¢); slight YES bias (56%); near-zero sell rate. **Why $p^* > p_m$:** the operator runs an external ensemble forecast (ECMWF/GFS) which is materially better than the crowd-derived p_m for short-horizon meteorological binaries. **Smallest profit pool** ($0.7 M aggregate) but 87% profitable.

4.10 R10. Strict HFT / market maker [S3]

Bet: *exactly* neutral (50% YES bias by construction for double-sided quoting); concentrated in the 20–55 ¢ mid-band where spread is widest; 0% sell rate (passive fills); median DCA 49 (re-quoting). **Why $p^* > p_m$:** the operator *is* the marginal counter-party at the best bid and best ask; their compensation is the half-spread. **Realised:** 77% profitable, median PnL $112 k—contradicting the popular claim that "HFT cannot make money on Polymarket". It can; it is just outscaled by the slow-information bucket.

4.11 R11. Diamond Hands hold-to-resolution [S1]

Bet: 0% sell rate by definition; mid-band heavy (45%). This is the *purest* S1 strategy: no microstructure exposure, no speculative timing. Buy once (or DCA a few times), wait. The operator's only edge is having $p^* - p_m - c > 0$ at the time of entry. **Empirical:** 76% profitable; median $51 k.

5 Aggregating Profits by Alpha Source

Figure 1 aggregates realised PnL across the regimes into the four alpha source buckets. **S1 (slow information) dominates**, accounting for roughly $140M of $177M realised panel PnL—driven entirely by Politics. S4 (cross-market) is second, primarily Sports modelling. S2 (behavioural) is moderate in absolute PnL but produces the highest median trader PnL, suggesting it is a more capital-efficient strategy. S3 (microstructure) is the smallest aggregate but the most distributionally even.

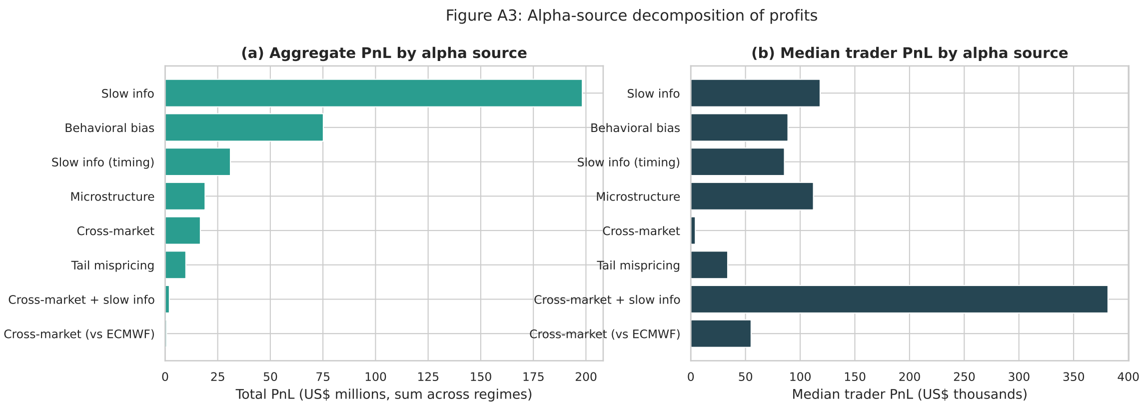

Figure 1: Aggregate realised PnL and median trader PnL by alpha source.

Figure 2 shows the price-band heatmap: each row is a regime; each column is the share of position size at that probability band. The patterns are sharp: **lottery is uniquely concentrated in the <20 ¢ band**; politics regimes (R1, R5) are uniquely concentrated in >55¢ HFT and contrarian sit in the mid-band. This is the operational fingerprint of the alpha-source mapping.

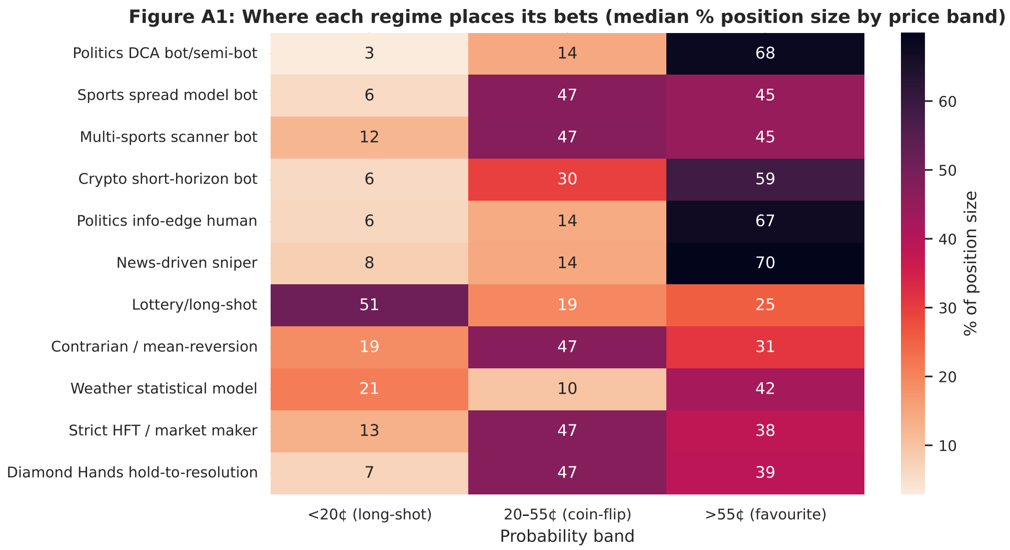

Figure 2: Price-band exposure by regime (median % of position size).

Figure 3 plots the median YES bias by regime. The two regimes hugging the 50% line—HFT (S3) and crypto short bot (S2)—are precisely the two regimes whose alpha source is *direction-agnostic*: market-making collects spread on either side, and tail-mispricing arbitrage harvests under-priced YES on both the up and down side of the same binary.

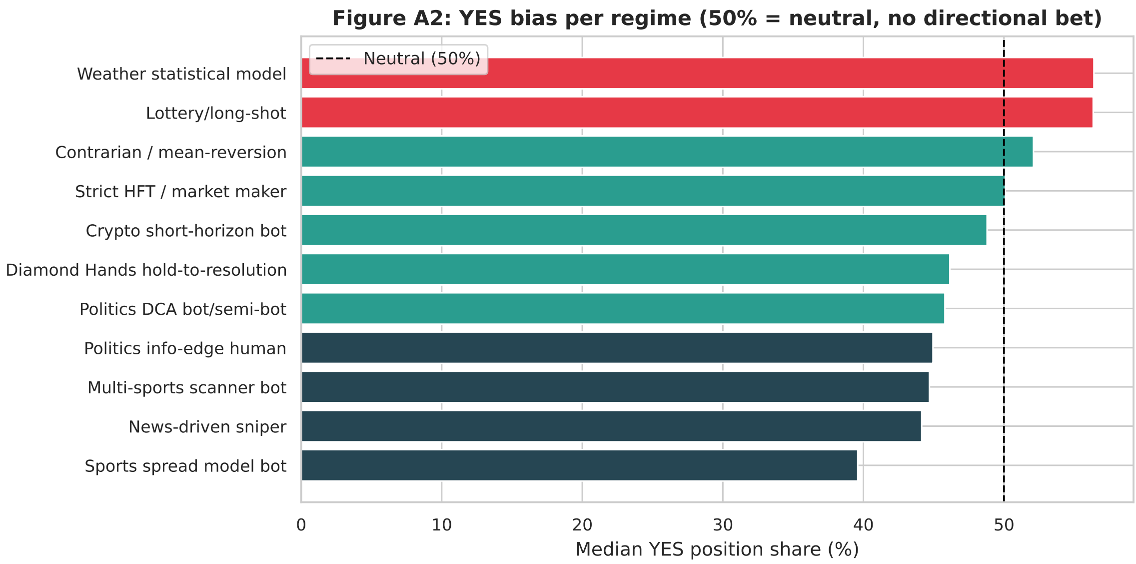

Figure 3: Median YES position share by regime; 50% = neutral.

Figure 4 plots the per-regime PnL distribution ordered by median. Politics DCA dominates the right tail by an order of magnitude, but the cross-regime spread is wider than the median suggests: every regime has both winning and losing operators, and within-regime dispersion exceeds between-regime dispersion for most buckets.

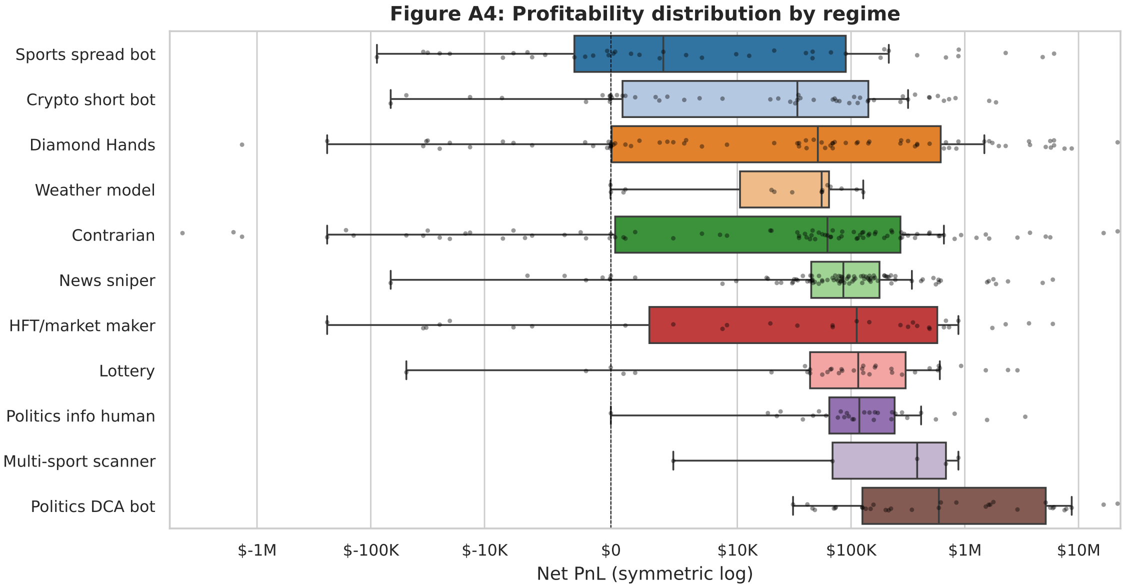

Figure 4: Per-regime PnL distribution (symmetric log).

6 Why This Matters

For practitioners. The alpha-source decomposition is a build/buy decision tree. If you can credibly forecast politics better than the marginal Polymarket retail counter-party, S1 is the largest pool of available alpha and the easiest to operationalise (DCA into a view, hold to resolution; no execution stack required). If you can build a fast statistical model of a domain (Vegas-quality sports lines, ECMWF-quality weather), S4 is more competitive but defensible. If you already own a market-making infrastructure, S3 has been demonstrated profitable in our panel and is essentially free of view risk. S2 is universally available but capacity-limited.

For researchers. The Polymarket dataset is uniquely useful because every alpha source is observable in trade-level wallet behaviour: probability-band exposure for S2, YES-bias for S3, sell-rate and DCA-density for S1. We have not found another asset class where the four classical alpha sources can be decomposed this cleanly from public data.

For exchange designers. If S1 is by far the dominant profit pool, the marginal Polymarket counter-party is a slow updater. A platform whose growth depends on retail liquidity should invest in *retail education about base rates* more than in microstructure optimisation; the current pattern transfers wealth from slow-updating retail to fast-updating professionals at scale.

What this is not. We do not claim that any of these regimes remains profitable in equilibrium. As more capital chases S1 in politics, p_m converges to p^* and the strategy decays. The point is descriptive, not prescriptive: in the snapshot we observe, all profits trace back to mispriced probabilities.

7 Conclusion

The Polymarket smart-money dataset reduces, after stripping the specifics of regimes, to a single sentence:

All profits on Polymarket are realised mispricings of implied probability.

The eleven regimes catalogued in Paper I are not eleven different games. They are eleven specific operational ways of locating $p^* \neq p_m$ in the implied-probability surface, executing trades that arbitrage the gap, and absorbing the microstructure cost. The sizes of the four alpha sources we identify—S1 slow information (largest by far), S4 cross-market (second), S2 behavioural (highest-median), S3 microstructure (smallest)—imply concrete allocations of effort for any operator entering this market. Polymarket is not a casino, not an HFT exchange, and not a venue for insider trading; it is a venue where positive expected value is the arithmetic difference between true and market-implied probabilities, and "smart money" is whoever measures that difference correctly and acts on it patiently enough to clear costs.

Companion paper. The descriptive-empirical companion, *Smart Money on Polymarket: A Behavioral Anatomy of 273 Top Prediction-Market Traders* (Paper I, same dataset and code release), contains the per-trader detail, the K-Means cluster derivation, and the full set of figures referenced here.

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